

Ethan Cavill

ethancavill@gmail.com | +1 858 518-4580 | +61 420 813 266 | U.S. / Australian Citizen

[linkedin](#) | [github](#) | [portfolio](#)

Experience

Generative AI Startup

Mar 2024 - Present

Lead Software Engineer

San Francisco, CA

- Led product engineering for a Laravel + Next.js **consumer AI product** with **200k+ users signed up**.
- Designed AI chat systems with **context compaction, memory extraction, and retrieval**.
- Deployed **GPU image/video pipelines** with ComfyUI workers, cloud uploads, and queue-based processing.
- Instrumented production reliability through **structured logging, Sentry, queue monitoring, and CI gates**.
- Reduced scaling risk through **queue prioritization**, slow query tuning, indexing, and caching critical requests.
- Managed cloud platform costs across **cloud functions, edge requests, media storage, and media optimization**.
- Owned **subscriptions, credits, payment webhooks, moderation, admin tooling**, framework upgrades, and dependency security hygiene.
- Led **engineering hiring**, assessing candidates across product, coding ability, and interpersonal skills.
- Accelerated DX with **strict linting, build checks, AI tooling harnesses, and automated PR code reviews**.

Tangible AI

Oct 2023 - Jan 2024

Machine Learning Engineer Intern

San Diego, CA

- Built open-source BERT **search engine with RAG**, achieving **88% improvement** in recall over TF-IDF engine.
- Presented information retrieval findings to **30+ attendees at San Diego Python User Group**.
- Proofread and **published edits to 'Natural Language Processing in Action.'**

Adelaide MRI

Jul 2021 - Oct 2023

Full Stack Software Engineer

Adelaide, SA

- Developed patented radiology reporting software with Angular, Flask, and ML; **boosting output 2-4x**.
- Automated invoicing with MediPass payment **API and webhook integration**.
- Wrote **integration/unit tests with PyTest (90%+ coverage)** and E2E tests with Cypress.
- Built patient booking system with React, GCP, and FastAPI; **reduced call burden by 50%**.
- **Managed large codebase version upgrades** and refactored business critical components.
- Led frontend migration from React to Angular, improving maintainability.
- Implemented strict linting/compiler settings; **reduced Sentry errors by 30%**.

Projects

deepworktimer.io | Next.js, FastAPI, PostgreSQL, Redis, Docker, AWS

2024 - Present

- Streak-based productivity system with **50+ weekly active users** through organic growth.
- Experimented with a **microservice-style architecture** using containerized FastAPI services.

ethancavill.me | Next.js, Tailwind CSS, Framer Motion, MDX, Vercel

2020

- Developed a portfolio website with React and Markdown to showcase projects, services, and written insights.

devarity.com | PHP, MySQL, Bootstrap, JQuery

2014 - 2015

- Developed an e-commerce platform **launching with 50+ titles** and partnerships with four indie publishing studios.

Skills

Languages

Python, JavaScript, TypeScript, Golang (Go), PHP, SQL, HTML/CSS, Bash

Technologies

React, Angular, Next.js, Redux, Zustand, RxJS, Docker, Linux, Flask, FastAPI, Django, Laravel, PostgreSQL, MongoDB, Redis, Celery, RabbitMQ, PyTorch, ComfyUI, PyTest, Cypress, Sentry, Horizon, Vercel, GCP, AWS

Concepts

Full-Stack Development, Frontend Development, Backend Development, Systems Design, Reactive Programming, State Management, Technical Writing, Agile, Scrum, Algorithms and Data Structures, Machine Learning, Data Science, Distributed Systems, Testing Strategy, CI/CD, DevOps, Observability, Queue Architecture, Performance Optimization, Cloud Cost Optimization

Education

Bachelor of Mathematical Sciences

Mar 2017 - Nov 2018

University of Adelaide

Adelaide, SA

- Coursework in linear algebra, calculus, and algorithms foundational to CS and ML.